Day 3: Switch and Router Configuration Basics

1. Introduction

Today we will learn two important networking concepts:

1. **VLAN**
2. **Static Routing**

We will use **Cisco Packet Tracer** for practical work.

Basic network:

PCs → Switch → Router → Router → Other Network

Part 1: VLAN

2. What is a VLAN?

VLAN means:

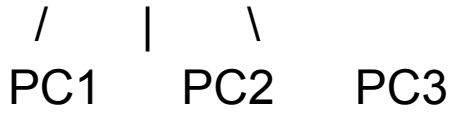
Virtual Local Area Network

A VLAN allows us to logically divide one physical switch into multiple separate networks.

Simple Example

Suppose a college has one switch:

Switch



PC1 and PC2 are for the **Computer Lab**.

PC3 is for the **Admin Department**.

Instead of buying separate physical switches, we can create VLANs:

VLAN 10 → Computer Lab
VLAN 20 → Admin

Now the switch can logically separate these devices.

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3. Why Do We Use VLANs?

VLANs are useful for:

- Network separation
- Security
- Department-wise organization
- Reducing broadcast domains
- Better network management

Real-Life College Example

A college may have:

VLAN 10 → Students

VLAN 20 → Faculty
VLAN 30 → Admin
VLAN 40 → Server

Even if all devices are connected to the same physical switch, VLANs logically separate them.

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4. VLAN Example

Suppose we have:

```
        Switch
      /   |   \
    PC1   PC2   PC3
```

Configure:

PC1 → VLAN 10
PC2 → VLAN 10
PC3 → VLAN 20

PC1 and PC2 belong to the same VLAN.

PC3 belongs to a different VLAN.

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5. Important VLAN Terms

VLAN ID

Every VLAN has a number.

Examples:

VLAN 10
VLAN 20
VLAN 30

Access Port

An access port normally connects an end device such as:

- PC
- Printer
- Server

Example:

PC1 → Switch Port Fa0/1

Fa0/1 can be configured as an access port for VLAN 10.

Trunk Port

A trunk port carries traffic for **multiple VLANs** between network devices.

Example:

Switch 1 ===== Trunk ===== Switch 2

or:

Switch ===== Trunk ===== Router

A trunk is commonly used when multiple VLANs need to travel across one physical link.

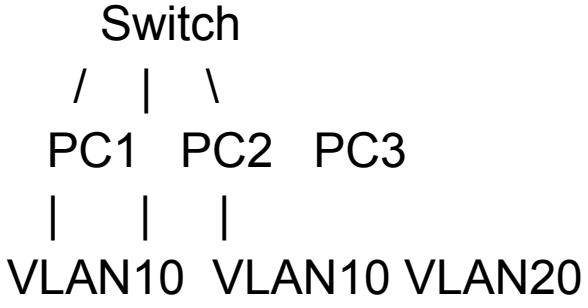
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6. VLAN Configuration Example

We will create:

VLAN 10 → LAB
VLAN 20 → ADMIN

Network:



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7. Cisco Packet Tracer Practical – Create VLANs

Step 1: Add Devices

Open Cisco Packet Tracer.

Add:

- 1 Switch
- 3 PCs

Use a switch such as:

2960

Connect:

PC1 → Switch Fa0/1
PC2 → Switch Fa0/2
PC3 → Switch Fa0/3

Use **Copper Straight-Through** cables.

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8. Step 2: Configure IP Addresses

PC1

IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0

PC2

IP Address: 192.168.10.20
Subnet Mask: 255.255.255.0

PC3

IP Address: 192.168.20.10
Subnet Mask: 255.255.255.0

For this basic VLAN practical, a default gateway is not required yet.

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9. Step 3: Open Switch CLI

Click the switch.

Go to:

CLI

Enter:

enable

Then:

configure terminal

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10. Step 4: Create VLAN 10

vlan 10
name LAB
exit

Now VLAN 10 is created.

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11. Step 5: Create VLAN 20

vlan 20
name ADMIN
exit

Now we have:

VLAN 10 → LAB
VLAN 20 → ADMIN

---

12. Step 6: Assign PC1 to VLAN 10

PC1 is connected to `FastEthernet0/1`.

Run:

interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit

---

13. Step 7: Assign PC2 to VLAN 10

interface fastEthernet 0/2
switchport mode access
switchport access vlan 10
exit

---

14. Step 8: Assign PC3 to VLAN 20

```
interface fastEthernet 0/3
switchport mode access
switchport access vlan 20
exit
```

---

15. Step 9: Check VLANs

Run:

```
show vlan brief
```

You should see something similar to:

```
VLAN Name                              Status    Ports
---- --------------------------------- --------
-----------------------------
1    default                           active    Fa0/4, Fa0/5...
10   LAB                               active    Fa0/1, Fa0/2
20   ADMIN                             active    Fa0/3
```

This confirms that the ports have been assigned to the correct VLANs.

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16. Test VLAN Communication

From PC1:

ping 192.168.10.20

PC1 and PC2 are in VLAN 10.

They should be able to communicate if their IP configuration and switch connections are correct.

Now from PC1:

ping 192.168.20.10

PC3 is in VLAN 20.

The ping should **not work with this basic Layer 2 configuration**, because VLAN 10 and VLAN 20 are separate networks and there is no Layer 3 routing configured between them.

This is an important concept:

VLAN 10
PC1 ↔ PC2
X
VLAN 20
PC3

To communicate between VLANs, we need **inter-VLAN routing** using a router or Layer 3 switch.

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17. Useful VLAN Commands

Display VLANs

show vlan brief

Display interfaces

show interfaces status

Display switch configuration

show running-config

Check a specific interface

show running-config interface fastEthernet 0/1

Part 2: Static Routing

18. What is Routing?

Routing means deciding **where an IP packet should go to reach another network**.

Example:

PC1
 |
Router 1
 |
Router 2
 |
PC2

PC1 and PC2 are on different networks.

The routers need routes to know where to forward packets.

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19. What is a Static Route?

A static route is a route that an administrator manually configures on a router.

Example:

Router 1
  |
  | 10.0.0.0/30
  |
Router 2

Router 1 can be manually told:

> "To reach Network 192.168.20.0/24, send packets to Router 2."

---

20. Static Routing Example

We will create this network:

PC1 -------- R1 -------- R2 -------- PC2

```
          |          |
      Network A    Network B
```

More specifically:

```
PC1
 |
192.168.10.0/24
 |
R1
 |
10.0.0.0/30
 |
R2
 |
192.168.20.0/24
 |
PC2
```

---

21. IP Address Plan

Device	Interface	IP Address	Subnet Mask
PC1	NIC	192.168.10.10	255.255.255.0

R1	G0/0	192.168.10.1	255.255.255.0
R1	G0/1	10.0.0.1	255.255.255.252
R2	G0/0	10.0.0.2	255.255.255.252
R2	G0/1	192.168.20.1	255.255.255.0
PC 2	NIC	192.168.20.10	255.255.255.0

The exact interface names may vary depending on the router model in Packet Tracer.

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22. Configure PC1

Set:

IP Address:
192.168.10.10

Subnet Mask:
255.255.255.0

Default Gateway:

192.168.10.1

---

23. Configure PC2

Set:

IP Address:
192.168.20.10

Subnet Mask:
255.255.255.0

Default Gateway:
192.168.20.1

---

24. Configure Router 1

Open R1 CLI.

enable
configure terminal

Configure the LAN interface.

interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

Configure the connection toward R2:

```
interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
```

---

25. Configure Router 2

Open R2 CLI.

```
enable
configure terminal
```

Configure the interface toward R1:

```
interface gigabitEthernet 0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
```

Configure the LAN interface:

```
interface gigabitEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

---

26. Check Interfaces

On R1:

show ip interface brief

On R2:

show ip interface brief

Look for:

Status: up
Protocol: up

If an interface is administratively down, check whether:

no shutdown

was configured.

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27. Check Connected Networks

On R1:

show ip route

You should see its directly connected networks, such as:

192.168.10.0/24
10.0.0.0/30

On R2:

show ip route

You should see:

10.0.0.0/30
192.168.20.0/24

But R1 does not automatically know how to reach `192.168.20.0/24` just because R2 knows about it.

This is where static routing is required.

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28. Configure Static Route on R1

On R1:

ip route 192.168.20.0 255.255.255.0 10.0.0.2

Meaning:

To reach:
192.168.20.0/24

Send traffic to:
10.0.0.2

---

29. Configure Static Route on R2

R2 also needs to know how to reach PC1's network.

On R2:

ip route 192.168.10.0 255.255.255.0 10.0.0.1

Meaning:

To reach:
192.168.10.0/24

Send traffic to:
10.0.0.1

Now both routers know how to reach the remote LAN.

---

30. Check Static Routes

On R1:

show ip route

You should see a static route similar to:

S   192.168.20.0/24 [1/0] via 10.0.0.2

S means **Static route**.

On R2:

show ip route

You should see:

S 192.168.10.0/24 [1/0] via 10.0.0.1

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31. Test Connectivity

First test PC1 → R1:

ping 192.168.10.1

Then PC1 → R2:

ping 10.0.0.2

Then the important test:

ping 192.168.20.10

If everything is configured correctly:

PC1 → R1 → R2 → PC2

The ping should succeed.

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32. Understand the Complete Packet Flow

PC1:

192.168.10.10

wants to communicate with:

192.168.20.10

The destination is on another network.

So PC1 sends the packet to its default gateway:

PC1
192.168.10.10
    |
    v
R1
192.168.10.1
    |
    v
10.0.0.2
R2
    |
    v
PC2
192.168.20.10

R1 checks its routing table and finds:

192.168.20.0/24 → 10.0.0.2

So R1 forwards the packet to R2.

R2 then forwards it to PC2.

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33. Important Difference: VLAN vs Routing

Students should clearly understand this.

VLAN

VLAN divides a switch into logical networks.

```
Switch
 |
 +--- VLAN 10
 |
 +--- VLAN 20
```

Routing

Routing connects different IP networks.

```
Network A
   |
 Router
   |
Network B
```

Together

```
VLAN 10 ----\
            Switch ---- Router ---- VLAN 20
VLAN 20 ----/
```

Routing is required when devices in different VLANs need to communicate.

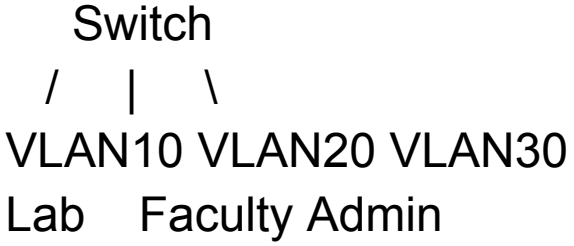
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34. Real-Time College Example

Suppose a college has:

VLAN 10 → Student Lab
VLAN 20 → Faculty
VLAN 30 → Admin

The switch separates these groups.



If the college wants Admin and Faculty networks to communicate with each other, a Layer 3 device must route traffic between those networks.

This is called **inter-VLAN routing**.

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35. Common Problems During Practical

Problem 1: Ping fails between PCs in same VLAN

Check:

1. IP address

2. Subnet mask
3. VLAN assignment
4. Cable
5. Switch port

Use:

show vlan brief

Problem 2: Router interface is down

Run:

show ip interface brief

If required:

interface gigabitEthernet 0/0
no shutdown

Problem 3: PC cannot reach another network

Check:

Default Gateway

Then check:

show ip route

Problem 4: Static route is missing

On R1:

show ip route

Check whether the remote network exists.

Example:

192.168.20.0/24

If not, configure:

ip route 192.168.20.0 255.255.255.0 10.0.0.2

Difference Between VLAN and Inter-VLAN

VLAN	Inter-VLAN
VLAN stands for Virtual Local Area Network.	Inter-VLAN means communication between different VLANs.
It divides one physical network into multiple logical networks.	It allows different VLANs to communicate with each other.

Devices in the same VLAN can communicate with each other, subject to normal network configuration.	Devices in different VLANs need Layer 3 routing to communicate.
It improves network organization and security.	It enables communication between separate network segments.
VLAN works mainly at Layer 2 (Data Link Layer).	Inter-VLAN routing works at Layer 3 (Network Layer).
Example: VLAN 10 for the HR department and VLAN 20 for the IT department.	Example: A computer in HR (VLAN 10) communicates with a server in IT (VLAN 20).

Simple Example

VLAN — Separate Networks

VLAN 10

HR — PC1, PC2

VLAN 20

IT — PC3, PC4

Devices are in separate logical networks.

Inter-VLAN — Communication Between Networks

Layer 3 Switch or Router

Routes traffic between VLAN 10 and VLAN 20

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36. Useful Cisco Commands

Students should practice these commands:

enable
configure terminal
show vlan brief
show ip interface brief
show ip route
show running-config
ping <IP-address>
traceroute <IP-address>

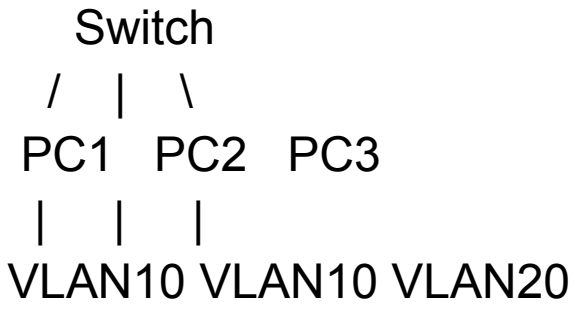
In Packet Tracer, the exact available commands can vary by device model and IOS version.

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37. Day 3 Practical Task

Part A – VLAN Task

Create this topology:



Configure:

VLAN 10 → LAB
VLAN 20 → ADMIN

Assign:

PC1 → 192.168.10.10
PC2 → 192.168.10.20
PC3 → 192.168.20.10

Test:

ping 192.168.10.20

Then:

ping 192.168.20.10

Students should explain why the first test and second test have different results in this basic Layer 2 setup.

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38. Part B – Static Routing Task

Create:

PC1 ---- R1 ---- R2 ---- PC2

Use:

PC1 = 192.168.10.10
R1 LAN = 192.168.10.1

R1-R2:
R1 = 10.0.0.1
R2 = 10.0.0.2

R2 LAN = 192.168.20.1
PC2 = 192.168.20.10

Configure static routes:

R1

ip route 192.168.20.0 255.255.255.0 10.0.0.2

R2

ip route 192.168.10.0 255.255.255.0 10.0.0.1

Test from PC1:

ping 192.168.20.10

Then test from PC2:

ping 192.168.10.10

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39. Student Documentation Task

Students should submit a small report containing:

VLAN Configuration

VLAN ID:
VLAN Name:
Switch Port:
Connected Device:
IP Address:

Static Routing

Router:
Destination Network:
Subnet Mask:
Next-Hop IP:

Example:

Router: R1

Destination: 192.168.20.0
Mask: 255.255.255.0
Next Hop: 10.0.0.2

Verification

Include the output/results of:

show vlan brief
show ip interface brief
show ip route
ping

Day 3 Learning Outcome

After this session, students should be able to:

- Explain what a VLAN is.
- Understand why VLANs are used.
- Create VLANs on a Cisco switch.
- Assign switch ports to VLANs.
- Understand access and trunk ports.
- Understand why different VLANs need routing to communicate.
- Explain what routing means.
- Explain static routing.
- Configure IP addresses on routers.
- Configure a static route.

- Verify routes using `show ip route`.
- Test connectivity using `ping`.
- Build a small routed network in Cisco Packet Tracer.